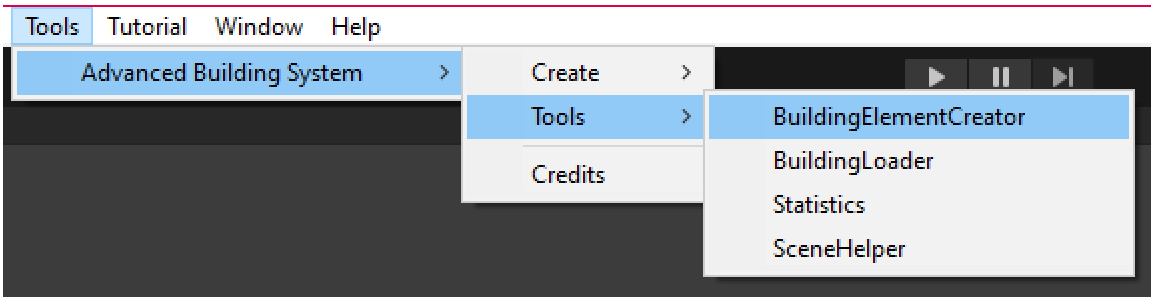
Tools

Building Element Creator

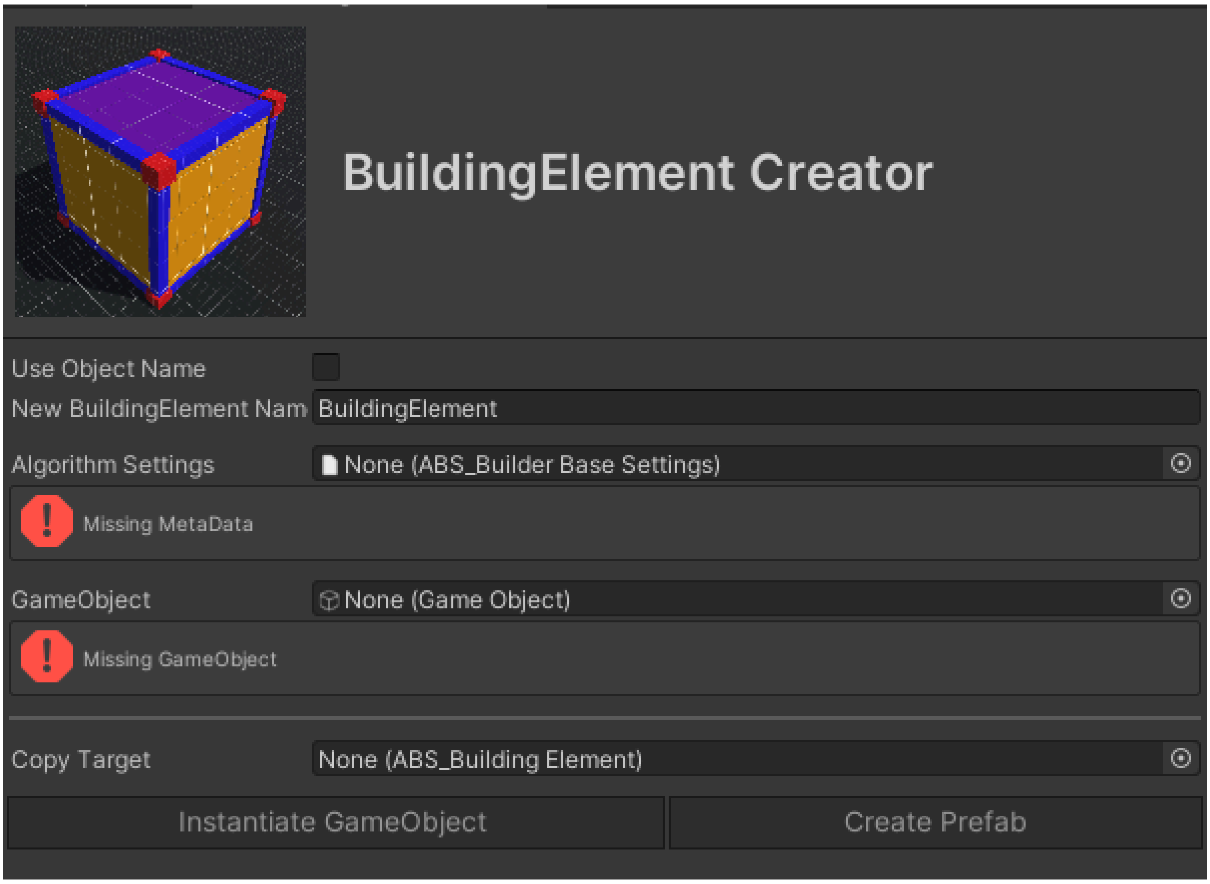
The ABS_BuildingElementCreator is a specific tool created for the Advanced Building System. The goal of the creator is to provide a tool that can be used to create ABS_BuildingElement. It is made to make the developers' workflow with creating the elements faster.
The creator can be found at:



How it is work?

The creator works in that way a GameObject has to be provided with the algorithm settings that should be used for that element and the creator will create a building element.
The creation process is the following:

1. Create the building element's GameObject
2. Instantiate the given GameObject as a child object to the new GameObject
3. Add a BoxCollider and add the "Layer Of Player" layer to the collider's include list.
    a. The layer is used from the given setting's layer collection.
4. Add the ABS_BuildingElement component
    a. Set the given settings
    b. Generate new PrefabGuid
    c. Refresh the BuildingElement's linkage which means that a ABS_BuildingElementLink component will added to every children GameObject what has a collider component
5. Copy the Copy Target's properties
6. Set the building element layer
    a. The new building element's GameObject's and every children's layer will be set to the "Layer Of BuildingElement" layer of the settings' Layer Collection



Properties

Use Object Name + New BuildingElement Name
If the “Use Object Name” boolean is true then the new building element’s name will be the provided “New BuildingElement Name” string. But if it is false then the given GameObject’s name will be used with a “_BuildingElement” postfix like:

```
m_BuildingElementName = $"{m_GameObject.name}_BuildingElement";
```

Algorithm Settings
An algorithm settings ScriptableObject should be provided that will be used during the creation process and also the BuildingElement will get it as a property.

GameObject
The given GameObject will be instantiated as a child for the new building element.

Copy Target
It can be null but if it is provided the given building element’s properties will be copied to the new building element.
The following properties will be copied:

- FinalElement
- PreBuilt
- Foundation
- Indestructible
- AreaType
- SnapToPreBuiltFinalElement

- ShouldAllowedByArea
- ShouldOverride
- DragBuildingEnabled
- DragBuildingBehaviour
- EnabledDragBuildingX
- EnabledDragBuildingZ
- HighlightCollection

Also the BoxCollider's size will be set to the Copy Target's Dimension Vector3

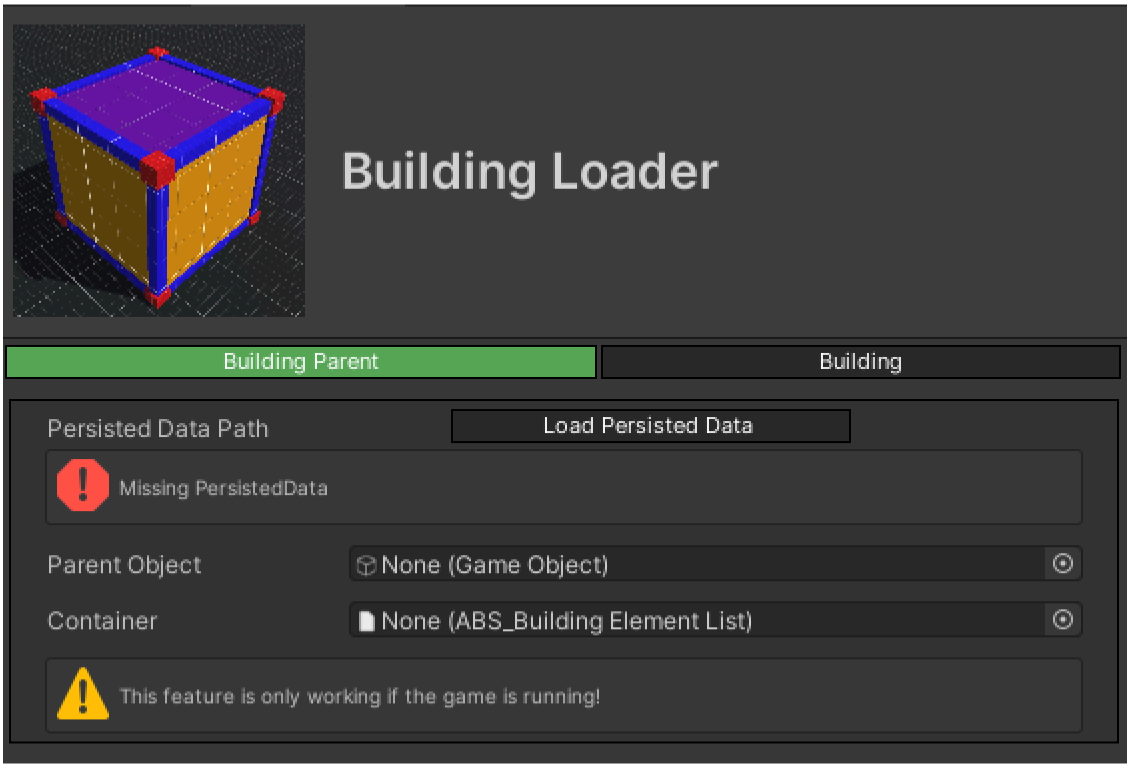
Instantiate

If every property has provided the following to options are available:

- Instantiate GameObject
  With this option the ABS_BuildingElement will be instantiated into the scene.
- Create Prefab
  With this option a prefab should be created to the selected path about the ABS_BuildingElement.
  Note that for this option the tool should instantiate the ABS_BuildingElement just like the other option but after the creation of the prefab the GameObject is removed immediately.

Building Loader

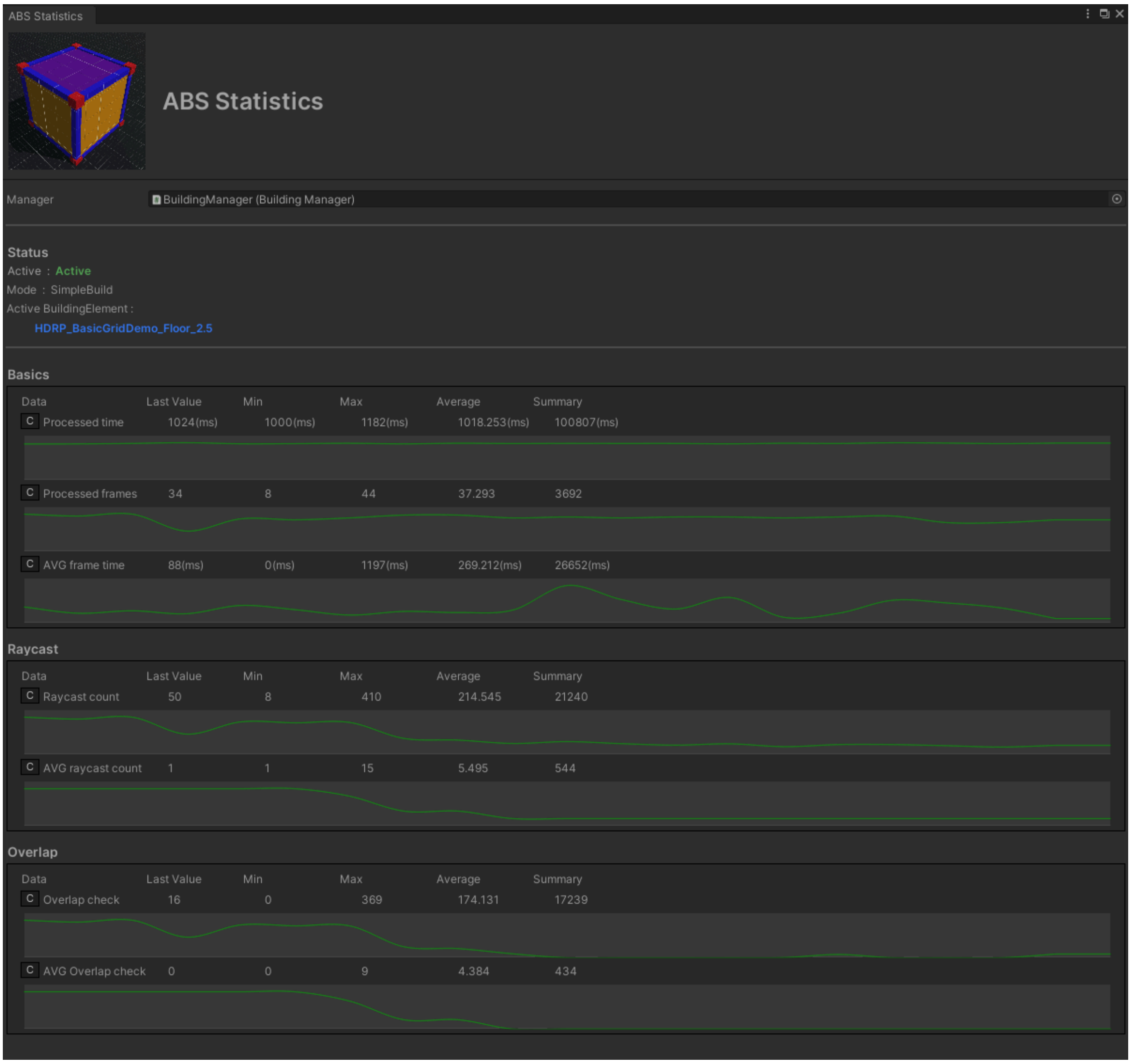
The BuildingLoader is a complementary tool for the persistence functionality of the BuildingManager. If you have a loading file for a Building/BuildingParent with this tool you can load it into your scene. Find it under the Tools > AdvancedBuildingSystem > Tools.
The tool works in that way you have to select a PersistedData file and a BuildingElementList (which contains every BuildingElement from the Buildings) should be provided and just press the Load button. It will load the persisted Building or BuildingParent into your scene. Also you can give a parent object from the Hierarchy .
Note that this tool can be used only when the game is running. If you do not want to run the scene this tool works totally fine with an empty scene.



ABS Statistics

The ABS Statistics is based on the IBuildingManagerStatisticsTracker. This tool gives a visual representation of the data of the Statistics feature.
This tool amplify the data of the Statistics with:

- Minimum Value
- Maximum Value
- Average Value
- Summary



Scene Helper

The purpose with the Scene Helper is helping the scene building process for the developers.

Analysis

This is currently the Scene Helper's only feature. It analyzes the active scene and identifies errors. If the tool can fix an error, it provides hotfix options.
The GameObject names are links on the UI that open the objects in the Inspector window.